



Rainwater Detector Project

**Third Year (Semester-VI) Bachelors in Information Technology,
Autonomous Program**

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Certificate

This is to certify that **Vanshika Ambwani(02), Gunjan Chandani(06) and Akruti Dabas(11)** have completed the project report on the topic **Rainwater Detector** satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Dr. Abhishek Chaudhari during the year 2024-2025 as prescribed by An Autonomous Institute Affiliated to University of Mumbai

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Abstract

Project Title: RAINWATER DETECTOR

In the modern era of smart automation, conventional protective methods are evolving into intelligent systems to enhance safety and convenience. This project presents a Rainwater Detection System that utilizes an Arduino Uno, rain sensor module, and automated shelter mechanism to detect rainfall and trigger a protective response for safeguarding personal belongings.

The primary objective of this system is to provide timely alerts about rainfall and automatically close the wardrobe shelter to prevent water damage to clothing and other stored items. Designed as a low-cost and offline solution, it requires no internet or GSM connectivity, making it ideal for homes, hostels, rural settings, and storage units where belongings are vulnerable to weather changes.

The rain sensor continuously monitors environmental conditions and, upon detecting rain, sends a signal to the microcontroller, which then activates a mechanical cover or alarm system. This ensures quick response to weather changes and enhances the protection of valuable items.

This project not only showcases a practical application of IoT and automation in daily life but also lays the groundwork for future enhancements such as mobile notifications, solar-powered operation, and integration with broader smart home systems. It emphasizes a blend of functionality, affordability, and user-centric design in modern smart living solutions.

Chapter 1

Introduction

1.1 Introduction

RAINWATER DETECTOR

In today's world of automation and smart living, there is a growing need for systems that can respond intelligently to environmental changes. One such useful innovation is the Rainwater Detector using Arduino Uno, which provides an effective solution to protect personal belongings from unexpected rainfall. This project is particularly designed to detect the presence of rain and trigger appropriate actions, such as alerting users or automatically closing a wardrobe shelter.

Using a rain sensor module connected to an Arduino Uno, the system continuously monitors weather conditions. When rainfall is detected, it sends an alert or activates a mechanism to prevent water from reaching items stored in wardrobes, balconies, or storage areas. This ensures timely action and protection without requiring manual intervention.

The project represents a simple yet impactful IoT-based automation system, ideal for homes, hostels, and rural areas where exposure to rainfall can lead to damage of clothes or other belongings. It promotes convenience, safety, and smart environmental responsiveness, while being cost-effective and easy to implement.

1.2 Objective

The main objective of this project is to:

- Design a cost-effective and automated rainwater detection system.
- Detect rainfall in real-time and trigger alerts or mechanical actions to protect stored items.
- Eliminate manual intervention by automating the process of closing a wardrobe shelter or cover.
- Enhance safety and convenience by preventing water damage to clothing and personal belongings.
- Promote the use of embedded systems and smart environmental sensing in day-to-day life, especially in homes, hostels, and rural areas.

Chapter 2

Review of Literature

2.1 Survey

With the advancement of embedded systems and environmental sensing, various rain detection systems have been developed to provide smart and automated responses to rainfall. These systems are primarily used to protect open-air installations, agricultural setups, and personal belongings. A brief overview of commonly used technologies includes:

- **Manual Rain Shelters:** Traditional methods involve manually covering or moving items during rainfall. While simple, they are not reliable or efficient in sudden weather changes.
- **GSM-Based Rain Alerts:** Some systems use GSM modules to send SMS alerts when rain is detected. Although effective for remote notifications, they depend on mobile networks and incur ongoing costs.
- **Wi-Fi and Cloud-Integrated Systems:** These systems provide real-time alerts and control via apps or dashboards. However, their reliance on continuous internet connectivity, routers, and cloud services makes them unsuitable for rural or low-connectivity areas.
- **Basic Alarm-Based Rain Detectors:** Many systems use simple rain sensors to trigger buzzers or alarms. While cost-effective, they do not provide automated mechanical responses or integration with smart shelters.
- **Smart Home Integration Modules:** Advanced systems can be integrated with home automation platforms like Alexa or Google Home. However, they involve complex programming, higher costs, and require smart home infrastructure.

Despite the availability of these technologies, challenges such as cost, complexity, power requirements, and dependence on internet or mobile networks still exist. There is a need for a low-cost, offline, and easy-to-deploy rainwater detection system that can work efficiently in diverse environments, especially in hostels, rural homes, and small-scale applications.

2.2 Research Gap

While various rain detection systems are available, several limitations persist in the mainstream solutions currently being used:

1. High Implementation and Maintenance Costs:

- GSM-based or IoT cloud-integrated systems incur recurring expenses such as SIM charges, data plans, or subscription fees for cloud services.
- Mechanized shelters integrated with smart home systems require additional actuators, smart controllers, and power backup, increasing the overall setup cost.

2. Dependence on Internet or Mobile Networks:

- Many advanced solutions depend on mobile data or Wi-Fi connectivity for remote alerts or app control, making them unsuitable for rural areas or places with weak signal coverage.

3. Lack of Automated Response Mechanisms:

- Most low-cost systems only provide an alarm or LED indication, but do not include automatic actions like shelter closure or cover deployment, which reduces their practical effectiveness.

4. Limited Accessibility for Non-Technical Users:

- Systems involving cloud dashboards, app setup, or complex configurations may be challenging to operate for users without technical backgrounds, such as elderly residents, students, or rural dwellers.

5. Neglect of Everyday Use Cases:

- Many solutions focus on agricultural or industrial applications and overlook simple domestic scenarios like protecting clothes in a wardrobe or items in a balcony, despite the high relevance of such use cases.

Chapter 3

Project Description

3.1 Problem Statement

Clothing and other personal belongings stored in open or semi-open areas such as balconies, rooftops, or near windows are often exposed to unexpected rainfall, leading to water damage and inconvenience. Traditional methods of protection rely heavily on manual intervention, which may not always be timely or practical—especially in the absence of individuals or during sudden weather changes.

While rain detection technologies exist, they often involve complex setups, internet dependency, or costly components, making them unsuitable for rural areas, hostels, or low-budget homes. Additionally, many affordable systems only provide basic alerts without taking automatic preventive action.

There is a need for a simple, low-cost, and offline rainwater detection system that can not only detect rainfall in real-time but also initiate a responsive mechanism, such as triggering an alarm or closing a protective shelter—ensuring the safety of stored items with minimal human effort.

This project addresses this challenge by developing an Arduino Uno-based Rainwater Detector that senses rainfall through a dedicated sensor and provides automatic alerts or physical response actions to prevent damage, especially in everyday domestic use cases.

3.2 Steps Involved

1. Component Collection and Setup

- Gather all required components: Arduino Uno, Rain Sensor Module (e.g., YL-83 or FC-37), Servo Motor (for shelter movement), resistors, jumper wires, and a half breadboard.
- Assemble the hardware on a breadboard, ensuring proper power supply and secure connections for all modules.

2. Circuit Design

- Connect:
 - Rain Sensor output pin to a digital input pin on the Arduino.
 - Servo motor control pin to a PWM-capable digital pin.
 - Power rails (VCC/GND) to components as per required voltage levels.
- Ensure current-limiting resistors are in place where necessary.
- Test individual components for correct pin mapping and responsiveness.

3. Arduino Programming

- Write code to:
 - Read analog/digital values from the rain sensor.
 - Control the servo motor to extend or retract the shelter based on rainfall detection.
 - Provide a fallback delay or rechecking mechanism for false positives.
- Upload the code to the Arduino Uno using the Arduino IDE.

4. System Behavior Setup

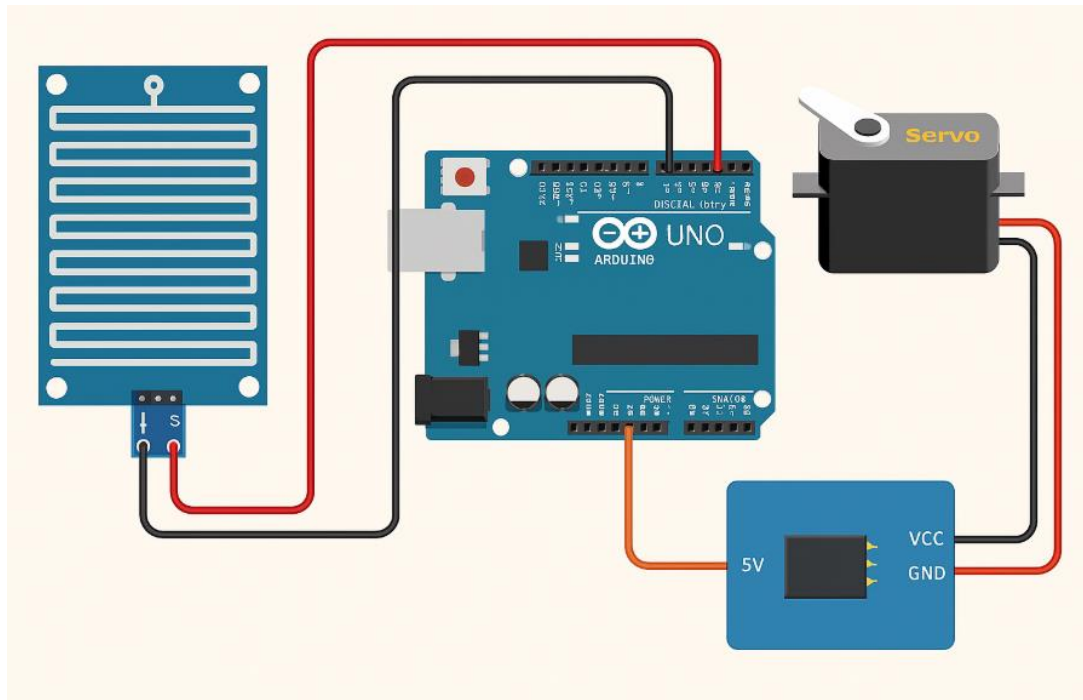
- Upload the final tested code to Arduino.
- Monitor the rain detection in real-time via the Serial Monitor.
- Ensure the system correctly identifies dry vs. rainy conditions.

5. Testing and Debugging

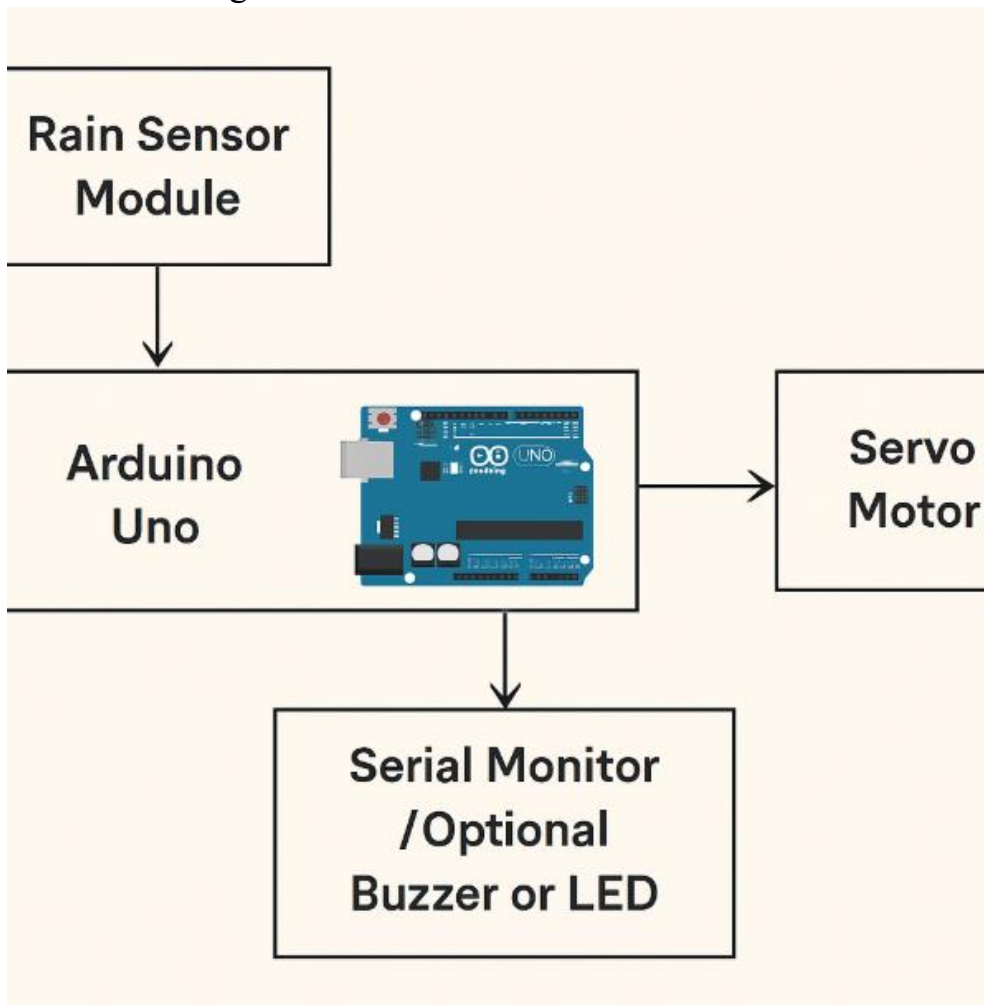
- Power the complete setup.
- Simulate rainfall (e.g., by gently wetting the sensor).
- Verify that the servo motor moves to cover the clothes/shelter upon detecting rain.
- Check that the servo retracts when no rain is detected.
- Adjust sensor sensitivity and motor response timing if needed.

6. Enclosure and Presentation

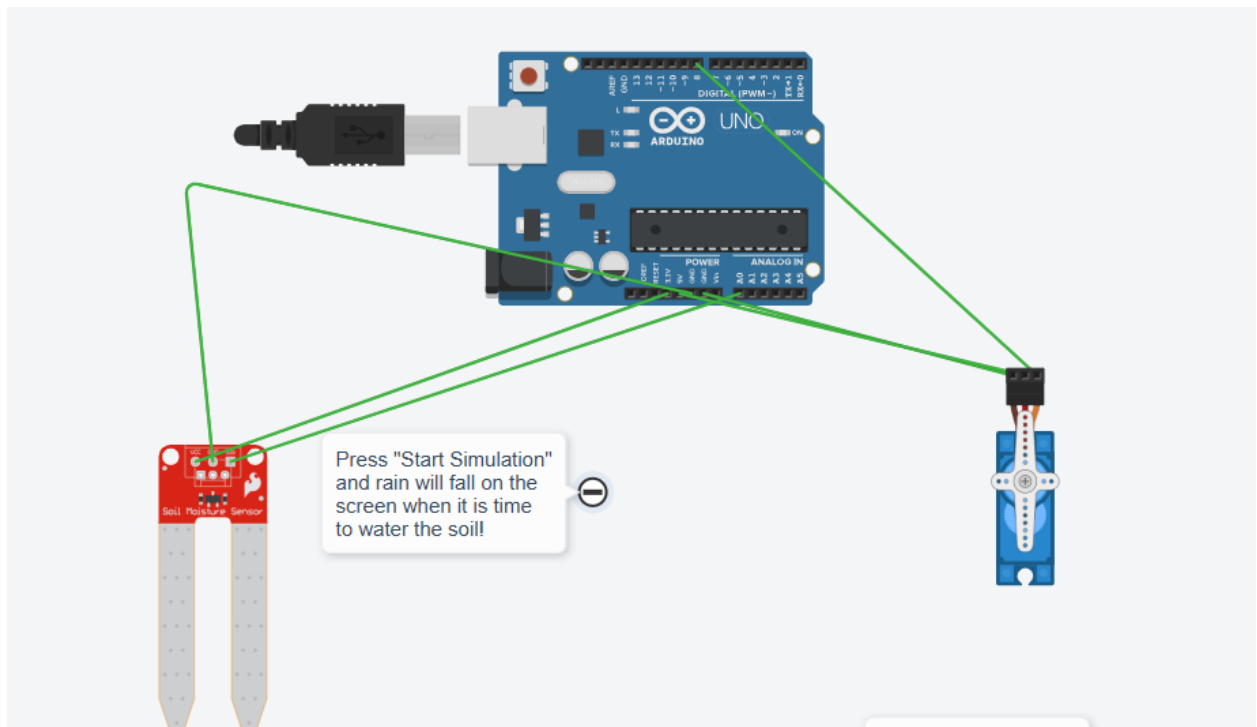
- Mount all components on a solid baseboard or enclosure.
- Label key parts for clarity during demonstration.
- Present a real-time scenario: Rain detected → Shelter covers clothes → Rain stops → Shelter retracts.
- Optionally, include an LED or buzzer for alert feedback.



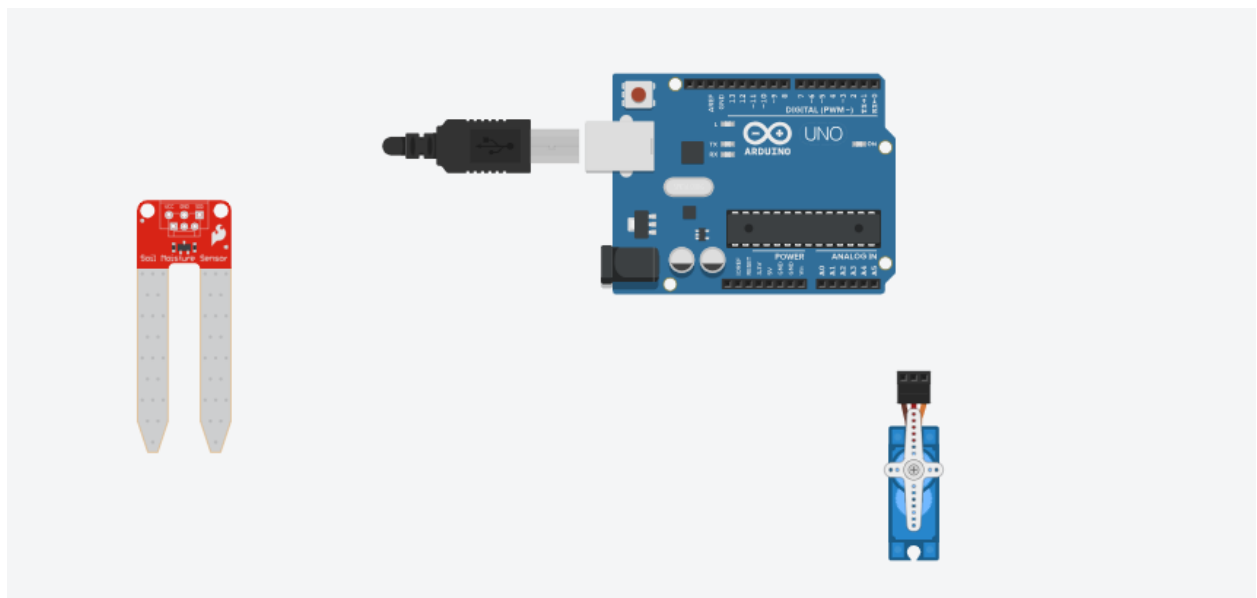
3.3 Circuit Diagram



3.4 Block Diagram



3.5 simulation diagram



3.6 component Description

3.4 Components Required (Rainwater Detector System)

1. Arduino Uno

- **Use:** Acts as the central controller to process input from the rain sensor and control the servo motor.
- **Description:** A microcontroller board based on the ATmega328P, ideal for beginner and intermediate IoT and automation projects.

2. Rainwater Sensor Module (Rain Drop Sensor)

- **Use:** Detects rain and sends a signal to the Arduino.
- **Description:** Consists of a rain-sensing plate and a signal conversion module. When raindrops fall on the plate, it alters the conductivity, which is detected and passed to Arduino via analog or digital pin.

3. Servo Motor (e.g., SG90 or MG996R)

- **Use:** Automatically moves the wardrobe cover to protect clothes from rain.
- **Description:** A position-controlled motor that rotates to specific angles based on PWM signals received from the Arduino.

4. Jumper Wires

- **Use:** Connect all components (rain sensor, servo, Arduino) for signal and power.
- **Description:** Flexible, plug-and-play wires with male/male ends, perfect for prototyping and easy circuit building.

5. Half Breadboard

- **Use:** Provides a platform for assembling the circuit without soldering.
- **Description:** A solderless board with multiple holes for easy and temporary connections of electronic components.

6. Power Supply (5V USB or Battery)

- **Use:** Supplies power to Arduino and connected modules.

Chapter 4

Implementation

4.1 Hardware

Component	Pin/Label	Connection
Arduino Uno	Vin	+ rail of breadboard
	GND	- rail of breadboard
	5V	Power other modules directly if needed
	D2	HC-05 TXD
	D3	HC-05 RXD (via voltage divider)
	D4	DHT11 Data Pin
	A4	LCD SDA
	A5	LCD SCL
	D5	↓ Rain sensor Digital Output (DO)

Rain Sensor Module (e.g., YL-83 or FC-37)

Pin	Connect To
VCC	+ rail of breadboard
GND	- rail of breadboard
DO (Digital Out)	D5 on Arduino

4.2 Software Code:

cpp

Copy

```
int rainSensorPin = 2;    // Pin connected to the rain sensor
int relayPin = 3;        // Pin connected to the relay
int rainStatus = 0;      // Variable to store the rain sensor value

void setup() {
    pinMode(rainSensorPin, INPUT); // Set rain sensor pin as input
    pinMode(relayPin, OUTPUT);     // Set relay pin as output
    Serial.begin(9600);           // Start serial communication for debugging
}

void loop() {
    rainStatus = digitalRead(rainSensorPin); // Read the rain sensor
    Serial.println(rainStatus);              // Print rain status to serial monitor

    if (rainStatus == HIGH) { // Rain detected
        digitalWrite(relayPin, HIGH); // Activate relay (close shelter)
        Serial.println("Rain detected. Sheltering...");
    } else {
        digitalWrite(relayPin, LOW); // Deactivate relay (open shelter)
        Serial.println("No rain. Shelter open.");
    }

    delay(1000); // Wait for 1 second before checking again
}
```

4.3 Algorithm:

Step 1: Start the program.

Step 2: Initialize the rain sensor and motor/servo controlling the rod.

Step 3: Continuously monitor the rain sensor's output.

Step 4: If rain is detected:

- Activate the motor/servo to **pull the rod** (move clothes inside shelter).
- Optionally, delay or wait until the rain stops.

Step 5: If no rain is detected:

- Activate the motor/servo to **push the rod** (move clothes outside for drying).

Step 6: Repeat steps 3–5 continuously.

Chapter 5

Results and Conclusion

Result

The implemented system successfully detects rainfall using the rain sensor and automatically moves the clothes-holding rod inside the shelter to prevent them from getting wet. Once the rain stops, the system moves the rod back outside.

The prototype responds accurately to rain detection and executes the rod movement via the motor without manual intervention. This ensures that clothes drying outdoors remain protected from unexpected rain, making the system efficient, reliable, and practical for daily use in households.

Conclusion

The **Rainwater Detection and Automatic Clothes Protection System** provides a smart and practical solution to a common household problem. By automatically detecting rainfall and moving clothes under shelter, the system eliminates the need for manual monitoring and intervention.

It combines simplicity with automation using basic sensors and Arduino, making it both cost-effective and easy to implement. This project not only demonstrates the potential of IoT in daily life but also encourages further development towards fully automated home solutions.

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